

Manual J Report



Prepared for Praestholm Residence (Rasmus)

643 Mt Pleasant Ave, West Orange, NJ 07052, USA

Created: Aug 11, 2026

Performed by: njenergyauditor@icloud.com

Calculations approved by ACCA to meet all Manual J 8v2.5 requirements

Heat: **54,260** Btuh

Cool: **28,485** Btuh

Heating Airflow: **1,653** CFM

Cooling Airflow: **1,176** CFM

Sensible Cooling: **25,739** Btuh

Latent Cooling: **2,747** Btuh

Sensible Ratio (SHR): **0.90**

Design Conditions

Weather Station: **ESSEX
COUNTY AP**

#724094, ASHRAE 2021

Latitude: 40.876
Elevation: 173 ft
Elevation Factor: 0.995

Indoor Heating 70°F

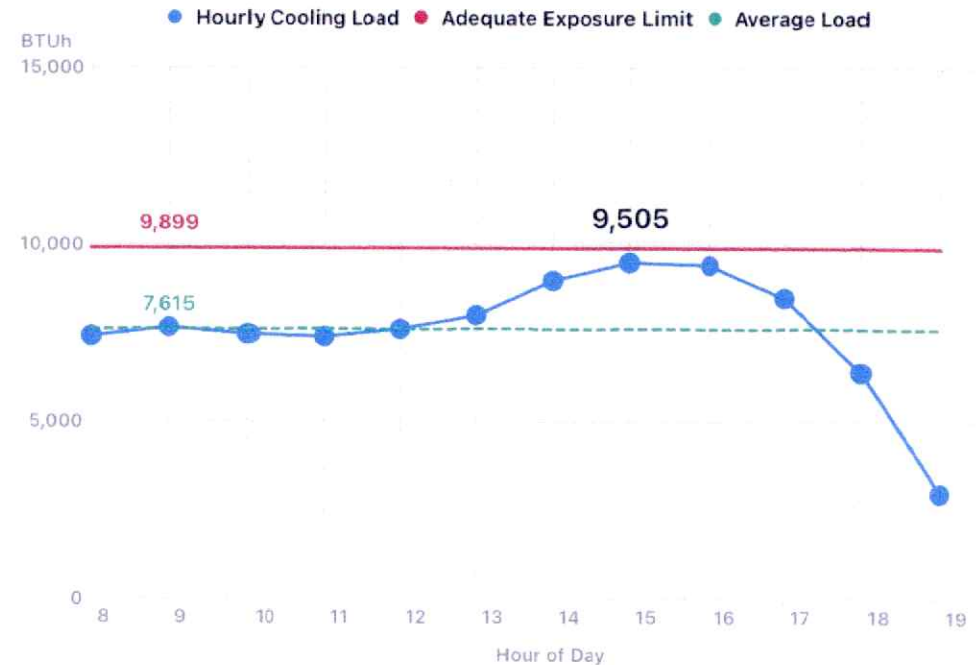
Indoor Cooling 75°F

Outdoor Heating 13°F

Outdoor Cooling 89°F

Design Grains 27.805

Daily Range: **Medium**



This application has adequate exposure diversity. The glass areas face a diverse set of directions and/or have sufficient shading (internal or external) to avoid solar gain peaks throughout the day.

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Progression A' AS-IS / tightness 'Loose'

Building Component Summary

Exposed Wall Area: 2,312 sq ft | Conditioned Above Grade Volume: 12,664 cu ft

Component	Details (sq ft)	Heating	Sensible Cooling	Latent Cooling
Floors	978	8,608	1,378	0
Above Grade Walls*	1,543	10,432	4,720	0
Below Grade Walls*	774	9,271	1,488	0
Ceilings*	1,547	5,008	4,406	0
Windows	152	5,554	5,339	0
Skylights	0	0	0	0
Doors (glass)	43	1,301	1,219	0
Doors (opaque)	39	932	428	0
Infiltration	Tightness: Loose, 2,261 sq ft ACH4 Cooling: 0.44 ACH4 Heating: 0.85	13,154	1,504	1,747
Internal	Occupants: 5 Plants: 0 Appliance Scenario: medium	0	3,550	1,000
Envelope Loads Sub Total		54,260	24,032	2,747
Ventilation	None	0	0	0
Ducts	—	0	0	0
Blower Motor Heat	—	0	1,707	0
AED Excursion	This application has adequate exposure diversity	0	0	0
Grand Total (Btuh)		54,260	25,739	2,747

*Net surface area accounting for doors, windows and skylights

Construction Details (Building Level)

Component	Description Construction ID	Qty	Net Area (sq ft)	Detail	HTM
Floors	Passive floor exposed to outdoor air - No insul, tile of vinyl 20P-0t	1	264.5	U: 0.521	29.49 5.21
	Bsmnt flr (> 2' below grade) - 32' short slab edge, no insul, any finish 21A-32	3	713.9	U: 0.020	1.13 0.00
Above Grade Walls	2x4 Wood Framed wall, Stucco/Siding - R-0 cavity/R-0 board 12A-0s w	3	247.9	U: 0.240	13.58 6.36
	2x4 Wood Framed wall, Stucco/Siding - R-11 cavity /R-0 board 12B-0s w	27	942.3	U: 0.097	5.49 2.34
	Garage Partition 2x4 Wood Framed wall, Stucco/Siding - R-11 cavity /R-0 board 12B-0s w	4	222.0	U: 0.097	5.49 3.08
	2x4 Wood Framed wall, Stucco/Siding - R-13 cavity /R-0 board 12C-0s w	2	130.5	U: 0.091	5.15 1.96
Below Grade Walls	Avg depth: 3 ft Bsmnt 8" Stone/Brick Wall - R-0 Board, No Framing, No Finish 15B0-0-x	2	186.8	U: 0.293	16.58 1.68
	Avg depth: 4 ft Bsmnt 8" Stone/Brick Wall - R-0 Board, No Framing, No Finish 15B0-0-x	2	215.4	U: 0.297	16.82 2.73
	Avg depth: 3 ft Bsmnt 8" Stone/Brick Wall - R-0 Board, Wd Frm R-11, Finished 15B11-0w-x	3	371.6	U: 0.088	4.98 0.68
Ceilings	Ceiling Under FHAvented w/o RB attic - R11 insul 16B-11	1	40.4	U: 0.081	4.58 4.03
	Ceiling Under FHAvented w/o RB attic - R13 insul 16B-13	3	541.2	U: 0.070	3.96 3.49
	Ceiling Under FHAvented w/o RB attic - R19 insul 16B-19	6	965.7	U: 0.049	2.77 2.44
Windows	Northeast Clear Single pane operable window/door 1A-c Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	2	8.3	U: 0.900 SHGC: 0.64	50.94 35.72
	Northwest Clear Single pane operable window/door 1A-c Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	1	9.8	U: 0.900 SHGC: 0.64	50.94 35.72
	Northwest Clear Single pane window w/storm 1C-c Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	1	9.1	U: 0.570 SHGC: 0.56	32.26 28.04
	Northeast LowE Double pane operable window/door 2A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	2	15.0	U: 0.550 SHGC: 0.56	31.13 28.40
	South LowE Double pane operable window/door 2A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	2	20.9	U: 0.550 SHGC: 0.56	31.13 20.72
	Southeast LowE Double pane operable window/door 2A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	3	26.6	U: 0.550 SHGC: 0.56	31.13 34.79
	Southwest LowE Double pane operable window/door 2A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	2	10.7	U: 0.550 SHGC: 0.56	31.13 34.79
	West LowE Double pane operable window/door 2A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	2	18.5	U: 0.550 SHGC: 0.56	31.13 40.55
	Southeast LowE Double pane operable window/door 3A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	1	6.8	U: 0.530 SHGC: 0.56	82.49 58.99
	Southwest LowE Double pane operable window/door 3A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	1	26.8	U: 0.530 SHGC: 0.56	34.50 39.67
Skylights	-	0	0	-	0 0
Doors (glass)	Northeast LowE Double pane operable window/door 3A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	1	43.4	U: 0.530 SHGC: 0.56	30.00 28.10
Doors (opaque)	Wood - Solid core w/metal storm 11F	1	21.5	U: 0.280	15.85 7.28
	Metal - Fiberglass core 11J	1	17.4	U: 0.600	33.96 15.60

● Heating HTM ● Cooling HTM



Below Grade

Basement Storage Side

Heat: 4,926 Btuh

Cool: 1,126 Btuh

Latent Cooling	288
Sensible Cooling	838
Sensible Ratio (SHR)	0.74
CFM Heating	150
CFM Cooling	38

Basement Utility Side

Heat: 9,988 Btuh

Cool: 1,938 Btuh

Latent Cooling	295
Sensible Cooling	1,643
Sensible Ratio (SHR)	0.85
CFM Heating	304
CFM Cooling	75

Basement Bathroom

Heat: 592 Btuh

Cool: 183 Btuh

Latent Cooling	15
Sensible Cooling	168
Sensible Ratio (SHR)	0.92
CFM Heating	18
CFM Cooling	8

1st Story

Exterior Front Picture

Heat: 4 Btuh

Cool: 3 Btuh

Latent Cooling	0
Sensible Cooling	3
Sensible Ratio (SHR)	1.00
CFM Heating	0
CFM Cooling	0

1F Bathroom

Heat: 620 Btuh

Cool: 343 Btuh

Latent Cooling	27
Sensible Cooling	316
Sensible Ratio (SHR)	0.92
CFM Heating	19
CFM Cooling	14

Front Corner Bedroom

Heat: 7,053 Btuh

Cool: 3,683 Btuh

Latent Cooling	290
Sensible Cooling	3,393
Sensible Ratio (SHR)	0.92
CFM Heating	215
CFM Cooling	155

Living Room & Foyer / Hallway

Heat: 3,952 Btuh

Cool: 2,652 Btuh

Latent Cooling	105
Sensible Cooling	2,547
Sensible Ratio (SHR)	0.96
CFM Heating	120
CFM Cooling	116

Rear Corner Bedroom

Heat: 2,910 Btuh

Cool: 1,722 Btuh

Latent Cooling	128
Sensible Cooling	1,593
Sensible Ratio (SHR)	0.93
CFM Heating	89
CFM Cooling	73

Kitchen & Dining

Heat: 14,819 Btuh

Cool: 5,475 Btuh

Latent Cooling	187
Sensible Cooling	5,288
Sensible Ratio (SHR)	0.97
CFM Heating	451
CFM Cooling	242

2nd Story

2/FA Cape Stairwell

Heat: 447 Btuh

Cool: 251 Btuh

Latent Cooling	18
Sensible Cooling	233
Sensible Ratio (SHR)	0.93
CFM Heating	14
CFM Cooling	11

Cape Bathroom

Heat: 585 Btuh

Cool: 325 Btuh

Latent Cooling	24
Sensible Cooling	301
Sensible Ratio (SHR)	0.93
CFM Heating	18
CFM Cooling	14

Cape Right Side Bedroom

Heat: 3,505 Btuh

Cool: 1,855 Btuh

Latent Cooling	159
Sensible Cooling	1,696
Sensible Ratio (SHR)	0.91
CFM Heating	107
CFM Cooling	77

Office, Etc

Heat: 4,859 Btuh

Cool: 2,673 Btuh

Latent Cooling	210
Sensible Cooling	2,462
Sensible Ratio (SHR)	0.92
CFM Heating	148
CFM Cooling	113

General

 **Stories** 2

 **Bedrooms** 4

 **Estimated Living Area** 1,604 sq ft

 **Year Built** 1946

 **Attached Garage: Open minutes per day**

 **Natural gas?** ☒

Existing heat pumps? ☐

 **Conditioned Basement?** ☒

Room over garage? ☐

 **HVAC System Type: Duc...**

Building Type: Single Family

Internal

 **Occupants** 5

Scenario: 2400 btuh - Standard Kitchen, utility and lighting

Heat: 0 Btuh
Cool: 4,550 Btuh

Latent Cooling 1,000
Sensible Cooling 3,550
Sensible Ratio (SHR) 0.78

Plants 12" or shorter 0

Plants 12" - 24" 0

Plants 24" or taller 0

Infiltration

Method: Tightness

 **Envelope Tightness: Loose**

Fireplaces 1

 **Infiltration Area** 2,261 sq ft

Building Type: Single Family

Heat: 13,154 Btuh
Cool: 3,250 Btuh

Latent Cooling 1,747
Sensible Cooling 1,504
Sensible Ratio (SHR) 0.46

Would measure ≥ 10 ACH50 via blower door test. Common for homes built pre-1950.

Scanned Living Area: 2,261 sq ft
Estimated Living Area: 1,604 sq ft

Ventilation

Type: None

Heat: 0 Btuh
Cool: 0 Btuh

Latent Cooling 0
Sensible Cooling 0
Sensible Ratio (SHR) 0.00